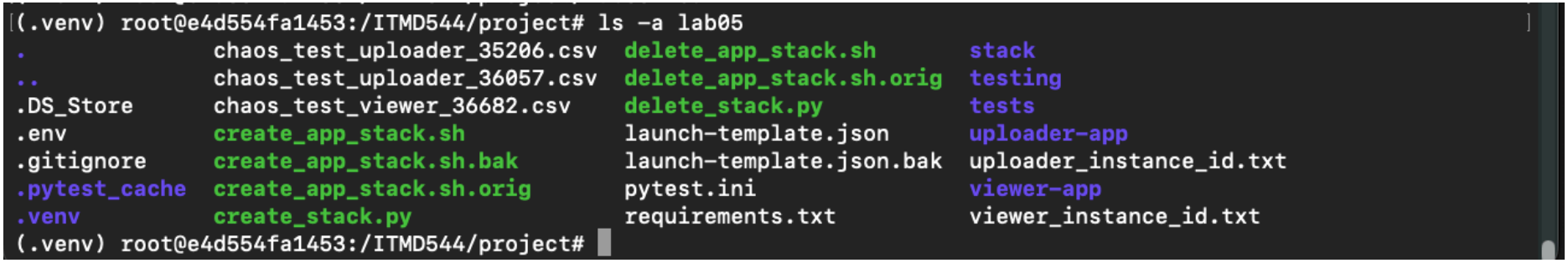
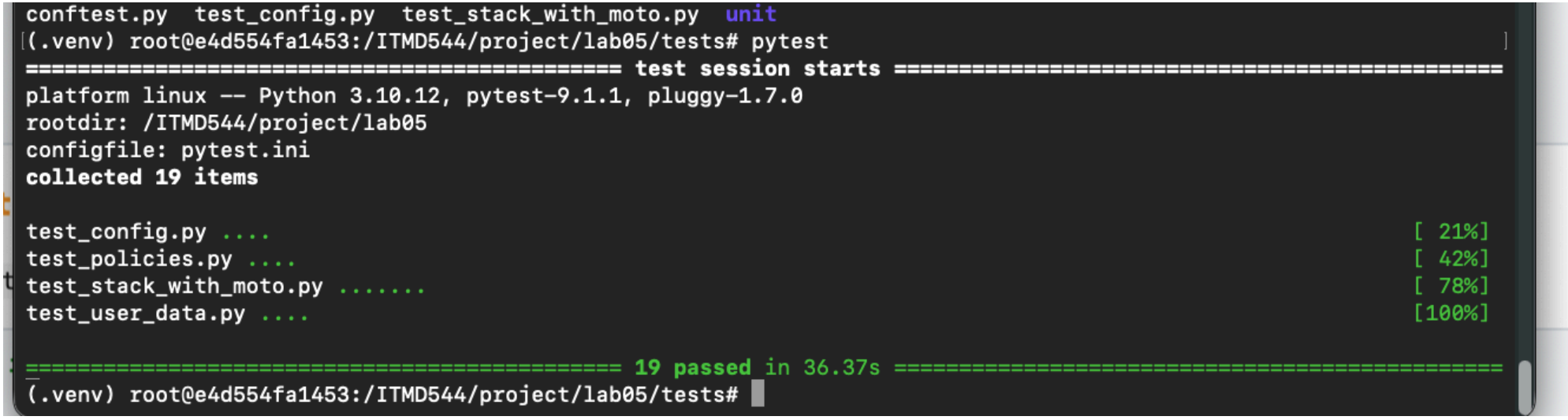
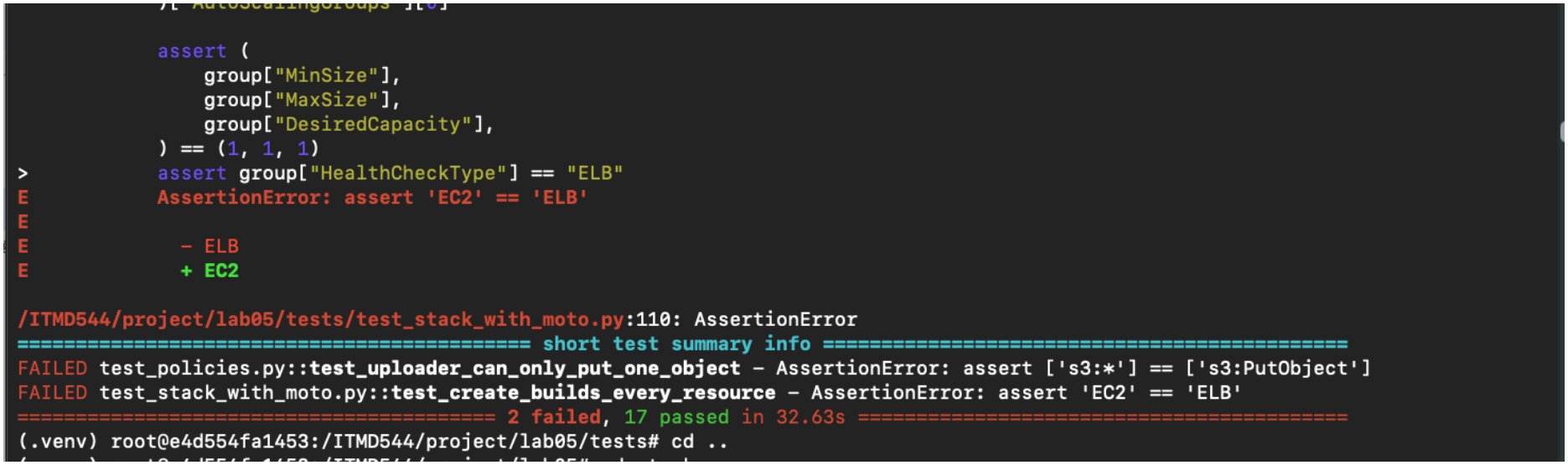
Output of `ls -a lab05` (or a screenshot) showing the layout from Part 0 and Section 4.



Screenshot of `pytest` showing **19 passed**.



Screenshot of one “break it on purpose” experiment: the changed line and the failing test message.


assert (
group["MinSize"],
group["MaxSize"],
group["DesiredCapacity"],
) == (1, 1, 1)
> assert group["HealthCheckType"] == "ELB"
E AssertionError: assert 'EC2' == 'ELB'
E
E - ELB
E + EC2
/ITMD544/project/lab05/tests/test_stack_with_moto.py:110: AssertionError
short test summary info
FAILED test_policies.py::test_uploader_can_only_put_one_object - AssertionError: assert ['s3:*'] == ['s3:PutObject']
FAILED test_stack_with_moto.py::test_create_builds_every_resource - AssertionError: assert 'EC2' == 'ELB'
2 failed, 17 passed in 32.63s
(.venv) root@e4d554fa1453:/ITMD544/project/lab05/tests# cd ..

```
collected 19 items

test_config.py ....                                                      [ 21%]
test_policies.py .F..                                                    [ 42%]
test_stack_with_moto.py F......                                          [ 78%]
test_user_data.py ....                                                   [100%]

================================================== FAILURES ==================================================
_____________________________________ test_uploader_can_only_put_one_object _____________________________________

    def test_uploader_can_only_put_one_object():
        policy = policy_for("uploader", BUCKET)

>       assert _allowed_actions(policy) == ["s3:PutObject"]
E       AssertionError: assert ['s3:*'] == ['s3:PutObject']
E
E         At index 0 diff: 's3:*' != 's3:PutObject'
E         Use -v to get more diff

test_policies.py:29: AssertionError
_____________________________________ test_create_builds_every_resource _____________________________________

env = (Settings(region='us-east-2', ami_id='ami-03cf127a', key_name='test-key', instance_type='t3.micro', bucket_name='unit-....Ela
sticLoadBalancingv2 object at 0x7f2c455d9b10>, autoscaling=<botocore.client.AutoScaling object at 0x7f2c453570a0>))

    def test_create_builds_every_resource(env):
```

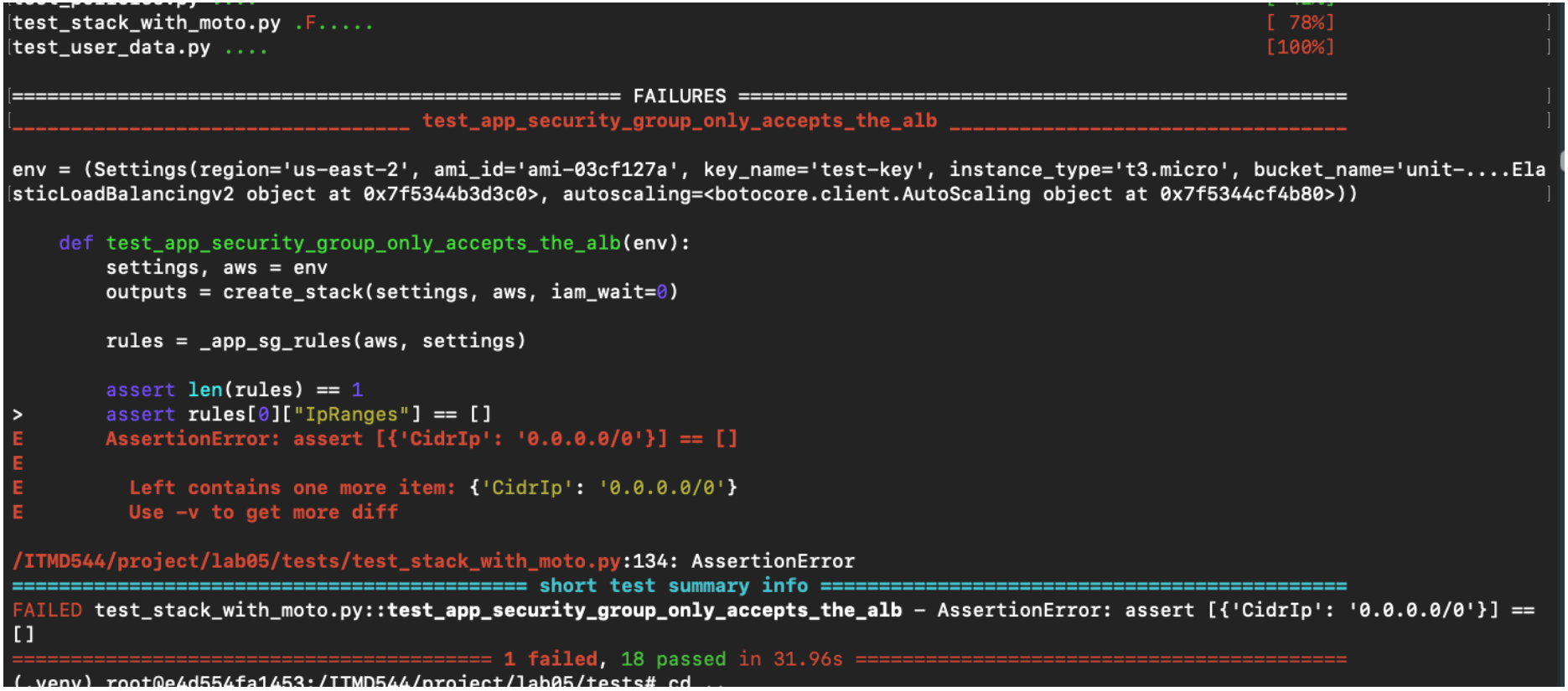
```
(.venv) root@e4d554fa1453:/ITMD544/project/lab05/tests# pytest
============================================= test session starts ==============================================
platform linux -- Python 3.10.12, pytest-9.1.1, pluggy-1.7.0
rootdir: /ITMD544/project/lab05
configfile: pytest.ini
collected 19 items
[                                                                                                              ]
[test_config.py ....                                                                                 [ 21%]    ]
[test_policies.py .F..                                                                               [ 42%]    ]
test_stack_with_moto.py F......                                                                      [ 78%]
test_user_data.py ....                                                                               [100%]
[                                                                                                              ]
================================================== FAILURES ==================================================
_____________________________________ test_uploader_can_only_put_one_object _____________________________________

    def test_uploader_can_only_put_one_object():
        policy = policy_for("uploader", BUCKET)

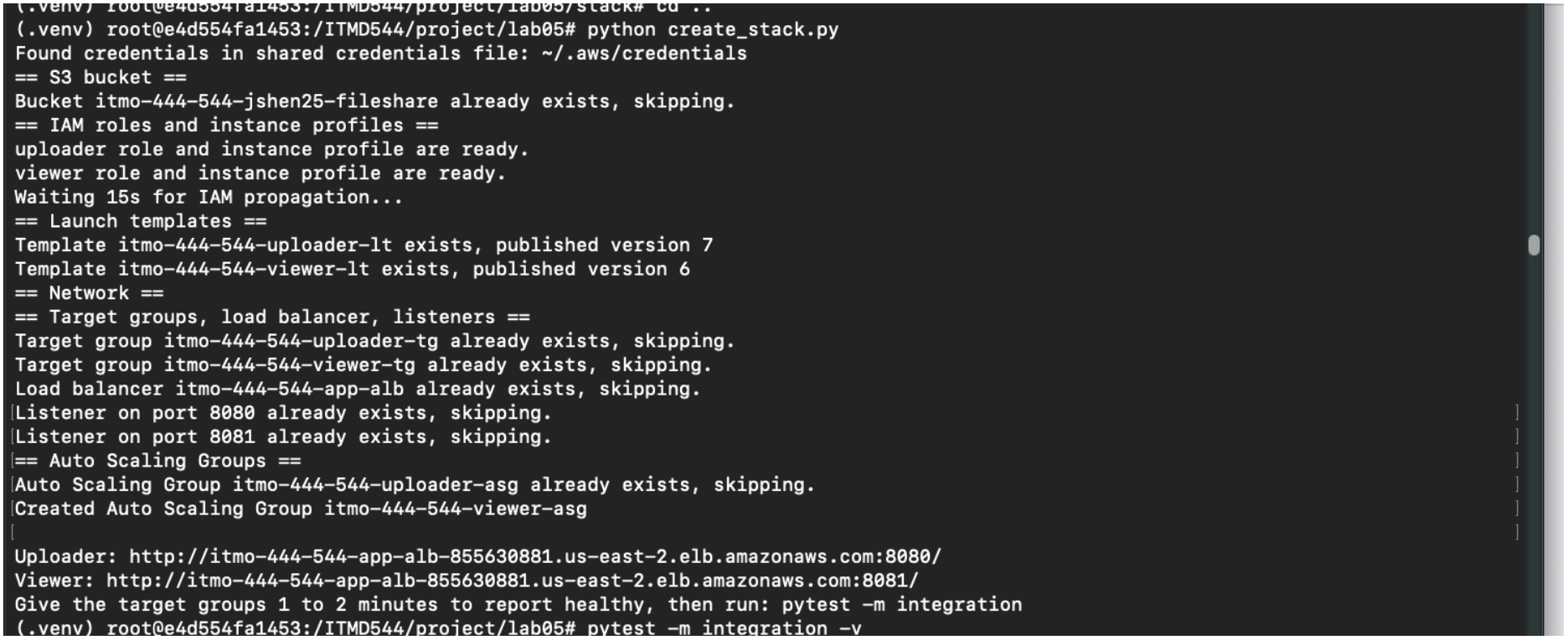
>       assert _allowed_actions(policy) == ["s3:PutObject"]
E       AssertionError: assert ['s3:*'] == ['s3:PutObject']
E
E         At index 0 diff: 's3:*' != 's3:PutObject'
E         Use -v to get more diff

test_policies.py:29: AssertionError
_____________________________________ test_create_builds_every_resource _____________________________________

env = (Settings(region='us-east-2', ami_id='ami-03cf127a', key_name='test-key', instance_type='t3.micro', bucket_name='unit-....Ela
sticLoadBalancingv2 object at 0x7efecff09b10>, autoscaling=<botocore.client.AutoScaling object at 0x7efecfc870a0>))
```



Screenshot of `python create_stack.py` completing, with both URLs.



Screenshot of `pytest -m integration -v` with all **14 tests passing**.

```
(.venv) root@e4d554fa1453:/ITMD544/project/lab05# pytest -m integration -v
============================================= test session starts ==============================================
platform linux -- Python 3.10.12, pytest-9.1.1, pluggy-1.7.0 -- /ITMD544/project/lab05/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /ITMD544/project/lab05
configfile: pytest.ini
testpaths: tests
collected 35 items / 21 deselected / 14 selected

tests/integration/test_application.py::test_uploader_serves_its_form PASSED                          [  7%]
tests/integration/test_application.py::test_viewer_serves_its_page PASSED                            [ 14%]
tests/integration/test_application.py::test_text_file_uploaded_on_one_tier_appears_on_the_other PASSED [ 21%]
tests/integration/test_application.py::test_non_txt_file_is_rejected PASSED                          [ 28%]
tests/integration/test_application.py::test_file_over_1mb_is_rejected PASSED                         [ 35%]
tests/integration/test_infrastructure.py::test_bucket_blocks_all_public_access PASSED                [ 42%]
tests/integration/test_infrastructure.py::test_uploader_role_can_only_write_the_shared_object PASSED [ 50%]
tests/integration/test_infrastructure.py::test_viewer_role_can_read_but_not_write PASSED             [ 57%]
tests/integration/test_infrastructure.py::test_load_balancer_is_public_and_spans_two_azs PASSED      [ 64%]
tests/integration/test_infrastructure.py::test_listeners_forward_to_the_right_target_groups PASSED   [ 71%]
tests/integration/test_infrastructure.py::test_target_groups_use_fast_health_checks PASSED           [ 78%]
tests/integration/test_infrastructure.py::test_auto_scaling_groups_are_configured_for_recovery PASSED [ 85%]
tests/integration/test_infrastructure.py::test_app_port_is_reachable_only_from_the_load_balancer PASSED [ 92%]
tests/integration/test_infrastructure.py::test_every_target_becomes_healthy PASSED                   [100%]

================================== 14 passed, 21 deselected in 5.39s ===================================
(.venv) root@e4d554fa1453:/ITMD544/project/lab05# pytest -v tests/integration/test_infrastructure.py::test_every_target_becomes_hea
lthy
============================================= test session starts ==============================================
```

Output of `pytest -m chaos -s` for both tiers, and the same run failing after you tightened the SLO.

```
        )

        assert saw_outage, (
            "the endpoint never went down, so there was nothing to recover from"
        )
        assert recovered_after is not None, "the endpoint did not come back in time"
>       assert recovered_after <= settings.recovery_slo_seconds
E       AssertionError: assert 79.85434889793396 <= 60
E        +  where 60 = Settings(region='us-east-2', ami_id='ami-0a1ff75d5f5ce322c', key_name='ITMO-444-544-ITMO-444-544-lab-keypair
', instanc...='itmo-444-544-viewer-asg', listener_port=8081, app_dir=PosixPath('/ITMD544/project/lab05/viewer-app')), app_port=3000
).recovery_slo_seconds

tests/integration/test_recovery.py:66: AssertionError
=========================================== short test summary info ============================================
FAILED tests/integration/test_recovery.py::test_tier_recovers_within_the_slo[uploader] - AssertionError: assert 83.14511847496033 <
= 60
FAILED tests/integration/test_recovery.py::test_tier_recovers_within_the_slo[viewer] - AssertionError: assert 79.85434889793396 <=
60
================================== 2 failed, 33 deselected in 165.29s (0:02:45) ====================================
(.venv) root@e4d554fa1453:/ITMD544/project/lab05# vi .env
```

```
        assert saw_outage, (
            "the endpoint never went down, so there was nothing to recover from"
        )
        assert recovered_after is not None, "the endpoint did not come back in time"
>       assert recovered_after <= settings.recovery_slo_seconds
E       AssertionError: assert 83.14511847496033 <= 60
E        +  where 60 = Settings(region='us-east-2', ami_id='ami-0a1ff75d5f5ce322c', key_name='ITMO-444-544-ITMO-444-544-lab-keypair
', instanc...='itmo-444-544-viewer-asg', listener_port=8081, app_dir=PosixPath('/ITMD544/project/lab05/viewer-app')), app_port=3000
).recovery_slo_seconds

tests/integration/test_recovery.py:66: AssertionError
_____________________________________ test_tier_recovers_within_the_slo[viewer] _____________________________________

aws = AwsClients(s3=<botocore.client.S3 object at 0x7f8555c81bd0>, iam=<botocore.client.IAM object at 0x7f85559835e0>, ec2=<...t.El
asticLoadBalancingv2 object at 0x7f85550192a0>, autoscaling=<botocore.client.AutoScaling object at 0x7f8554f8e830>)
settings = Settings(region='us-east-2', ami_id='ami-0a1ff75d5f5ce322c', key_name='ITMO-444-544-ITMO-444-544-lab-keypair', instanc..
.='itmo-444-544-viewer-asg', listener_port=8081, app_dir=PosixPath('/ITMD544/project/lab05/viewer-app')), app_port=3000)
outputs = StackOutputs(vpc_id='vpc-02561a587b9fb2f34', subnet_a='subnet-0c7e269d21fe070c9', subnet_b='subnet-0647ca4a3f6d5884f',...
iewer_tg_arn='arn:aws:elasticloadbalancing:us-east-2:786589010080:targetgroup/itmo-444-544-viewer-tg/c2f2c95b81276673')
tier_name = 'viewer'

    @pytest.mark.parametrize("tier_name", ["uploader", "viewer"])
    def test_tier_recovers_within_the_slo(aws, settings, outputs, tier_name):
        tier = settings.tier(tier_name)
        url = tier_url(outputs, settings, tier_name)

        assert probe(url), "the endpoint must be healthy before the experiment starts"
```

```
(.venv) root@e4d554fa1453:/ITMD544/project/lab05# pytest -m chaos -s
========================================== test session starts ==========================================
platform linux -- Python 3.10.12, pytest-9.1.1, pluggy-1.7.0
rootdir: /ITMD544/project/lab05
configfile: pytest.ini
testpaths: tests
collected 35 items / 33 deselected / 2 selected

tests/integration/test_recovery.py
uploader: terminated i-05cedbf9621edf715, recovery took 83.14511847496033s (SLO 60s)
F
viewer: terminated i-0b8343039acbb0abb, recovery took 79.85434889793396s (SLO 60s)
F

=============================================== FAILURES ================================================
_____________________________ test_tier_recovers_within_the_slo[uploader] ______________________________

aws = AwsClients(s3=<botocore.client.S3 object at 0x7f8555c81bd0>, iam=<botocore.client.IAM object at 0x7f85559835e0>, ec2=<...t.El
asticLoadBalancingv2 object at 0x7f85550192a0>, autoscaling=<botocore.client.AutoScaling object at 0x7f8554f8e830>)
settings = Settings(region='us-east-2', ami_id='ami-0a1ff75d5f5ce322c', key_name='ITMO-444-544-ITMO-444-544-lab-keypair', instanc..
.='itmo-444-544-viewer-asg', listener_port=8081, app_dir=PosixPath('/ITMD544/project/lab05/viewer-app')), app_port=3000)
outputs = StackOutputs(vpc_id='vpc-02561a587b9fb2f34', subnet_a='subnet-0c7e269d21fe070c9', subnet_b='subnet-0647ca4a3f6d5884f',...
iewer_tg_arn='arn:aws:elasticloadbalancing:us-east-2:786589010080:targetgroup/itmo-444-544-viewer-tg/c2f2c95b81276673')
tier_name = 'uploader'

    @pytest.mark.parametrize("tier_name", ["uploader", "viewer"])
    def test_tier_recovers_within_the_slo(aws, settings, outputs, tier_name):
        tier = settings.tier(tier_name)
        url = tier_url(outputs, settings, tier_name)

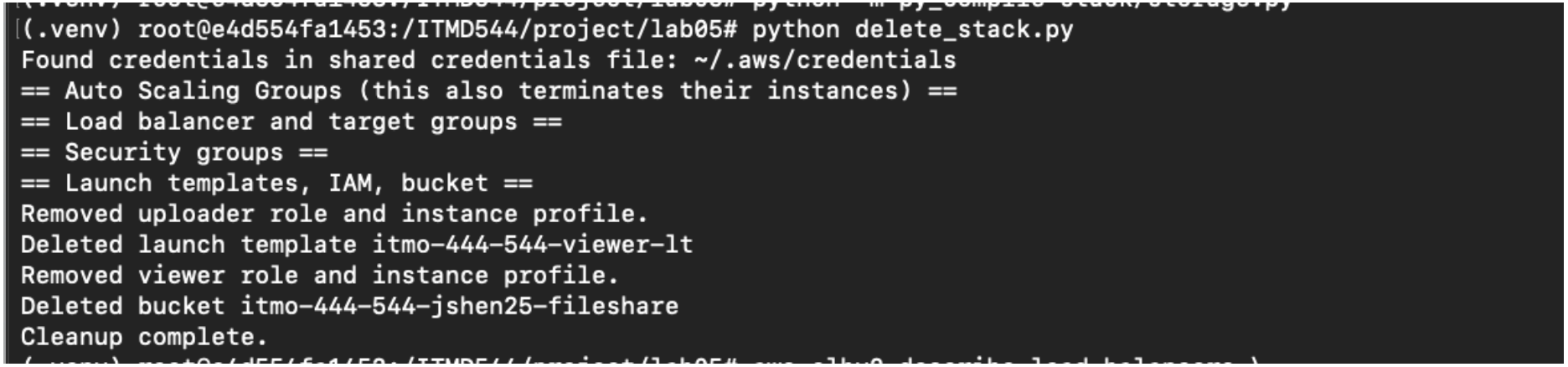
        assert probe(url), "the endpoint must be healthy before the experiment starts"

        victim = _in_service_instance(aws, tier.asg_name)

        assert victim, f"no in-service instance in {tier.asg_name}"

        aws.autoscaling.terminate_instance_in_auto_scaling_group(
            InstanceId=victim,
```

Screenshot of `python delete_stack.py` completing, and a second run completing with no errors.



One paragraph on what Moto proved, what it could not, and which tests cover the gap.

Moto showed that our stack creation and deletion code worked against simulated AWS services, including resource configuration and repeated cleanup. However, it could not prove that real EC2 instances would start correctly, IAM permissions would work in practice, the applications would communicate through the load balancer, or recovery would meet our SLO. The infrastructure integration tests check the deployed AWS configuration and target health, while the application integration tests check page responses, file uploads, cross-tier access, and invalid-file rejection. The chaos tests cover recovery by terminating a real instance in each tier and measuring whether service returns within the configured SLO.

One paragraph comparing `ignore_errors("EntityAlreadyExists")` with `2>/dev/null || true`.

`ignore_errors("EntityAlreadyExists")` ignores only the expected AWS error when a resource already exists, allowing the script to run again while still reporting other failures, such as missing permissions or invalid settings. In contrast, `2>/dev/null || true` hides all

standard error output and makes a failed command appear successful, potentially hiding serious problems and allowing the script to continue with missing resources. Therefore, catching a specific error makes automation easier to debug and more reliable.

Your answer to Challenge 5.
N/A
no time to do.